

INVOLVEMENT OF METHYL FARNESOATE IN THE REGULATION OF MOLTING AND REPRODUCTION IN THE FRESHWATER CRAB *OZIOTELPHUSA SENEX SENEX*

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A B S T R A C T

The influence of methyl farnesoate (MF) in the regulation of molt and gonad development in the crab *Oziotelphusa senex senex* was investigated. Injection of methyl farnesoate into female and male crabs significantly ($P < 0.0001$) increased mean oocyte diameter (236.37%) and testicular follicle diameter (25.72%) as well as mean gonad indices (females 1304.35%; males 38.00%) and also accelerated the molting (females 80%; males 100%). These results provide strong evidence that methyl farnesoate is involved in the control of both molting and reproduction in crabs.

Regulation of molting and reproduction in crustaceans involves the steroidal molting hormones, the ecdysteroids (Chang, 1993; see review Subramoniam, 2000), and the sesquiterpenoid methyl farnesoate (MF) (Laufer *et al.*, 1998). The precursor of 20-hydroxy ecdysteroids is synthesized and secreted from the Y-organ (Gabe, 1953). This process is under the negative control of molt inhibiting hormone (MIH) (see review by Webster, 1998). Likewise, MF is synthesized and secreted from mandibular organs (MOs) of crustaceans and is also under the negative control of mandibular organ inhibiting peptide hormone (MOIH), derived from the X-organ–sinus gland complex in the eyestalk (Laufer *et al.*, 1987a; Wainwright *et al.*, 1996; Liu *et al.*, 1997).

The MOs were first described by Le Roux (1968) and often confused with the Y-organs, because of the close anatomical proximity and the histological similarities. Both the organs show ultrastructural and cyclical changes correlated with the molt stage of the animal (Auto *et al.*, 1974). The MOs can now be conveniently distinguished from Y-organs biochemically, because the former synthesize and secrete MF and later ecdysteroids. Methyl farnesoate is chemically similar to the insect juvenile hormone III (Laufer *et al.*, 1987b). The physiological role of this compound has been subject to debate in recent years (Laufer *et al.*, 2002; Olmstead and LeBlanc, 2002). Methyl farnesoate is known to be involved in the regulation of molting (Yudin *et al.*, 1980; Chang *et al.*, 1993; Tamone and Chang, 1993; Wilder *et al.*, 1995), some aspects

of reproduction (Borst *et al.*, 1987; Laufer and Biggers, 2001; Sagi *et al.*, 1993; Reddy and Ramamurthi, 1998; Kalavathy *et al.*, 1999), morphogenesis (Laufer *et al.*, 1997; Laufer and Biggers, 2001; Abdu *et al.*, 1998; Rotllant *et al.*, 2000) and general protein synthesis (Paulson and Skinner, 1988).

A clear knowledge about the molt and reproduction stages of edible crustaceans will be useful in hatchery and farming operations. The common practice of inducing precocious reproduction in Crustacea is by eyestalk extirpation. This procedure, though effective in inducing reproduction in many commercially important crustacean species, frequently kills the animals either at the time of surgery or after the operation and leads to permanent damage. This operation frequently results in inferior quality and less quantity of seed (Choy, 1987). Consequently, finding ways to stimulate gonad development and maturation by nonsurgical procedures is a major goal of several laboratories. An elaborate programme to induce molting and reproduction in selected edible crustaceans has been undertaken in this laboratory. We have demonstrated stimulation of ovarian growth in the crab after Leucine-enkephalin injection (Kishori and Reddy, 2000). The present report examines the effect of injection of MF on molt and reproduction in the fresh water edible rice field crab, *Oziotelphusa senex senex* Fabricius.

MATERIALS AND METHODS

Indian rice field crabs, *Oziotelphusa senex senex* Fabricius, were collected from paddy fields in and around Tirupati

Table 1. Effect of injection of methyl farnesoate on molting and reproduction in female crab, *Oziotelphusa senex senex*.

	Gonad stage	Color of the ovary	Ovarian index	Oocyte diameter (μm)	Molt stage
Control ($n = 25$)	Previtellogenic	White	0.138 ± 0.03	22.74 ± 3.00	Intermolt
Concurrent control ($n = 25$)	Previtellogenic	White	0.113 ± 0.01 (-18.11)	23.92 ± 4.04 (5.18)	Intermolt (100)
MF injected ($n = 25$)	Vitellogenic I ($n = 8$)	Pale yellow	0.178 ± 0.04 (28.99)	20.41 ± 3.03 (-10.25)	Premolt (32)
<i>F</i> ratio			18.348	3.309	
<i>P</i> value			< 0.0001	= 0.0456	
	Vitellogenic II ($n = 3$)	Orange	0.306 ± 0.12 (121.74)	42.96 ± 9.40 (88.92)	Intermolt (4)
<i>F</i> ratio			40.658	33.117	Premolt (8)
<i>P</i> value			< 0.0001	< 0.0001	
	Vitellogenic III ($n = 10$)	Brown	1.938 ± 0.16 (1304.35)	76.49 ± 7.04 (236.37)	Intermolt (16)
<i>F</i> ratio			2481.1	544.50	Premolt (24)
<i>P</i> value			0.0001	< 0.0001	Moulted (16)

Values are mean $\pm$ S.D.

Values in parentheses are percentage change from control crabs.

(Andhra Pradesh, India) and maintained in the laboratory at $28^\circ \pm 1^\circ\text{C}$ in freshwater tubs. They were acclimatized to laboratory conditions (12:12 L:D) for one week before being used in experiments. The animals were fed with sheep meat *ad libitum* daily, and the medium was changed 3 h after feeding. Only intermolt (stage C_4) crabs were used to study the effect of MF on molting and reproduction. All the experiments were conducted during April–June.

Trans, trans-methyl farnesoate was dissolved in 95% ethanol, which was the solvent vehicle. One hundred fifty crabs were divided into six groups of 25 animals each. Groups with odd numbers were females, and those with even numbers were males. Groups 1 and 2 served as initial controls, and crabs in these groups received no treatment and were sacrificed on the first day of the experiment. Crabs in groups 3 and 4 received 10 μL of 10% ethanol through the arthrodial membrane of the coxa of the third pair of walking legs and served as concurrent controls. Crabs in groups 5 and 6 received MF at a dose of 15 ng/crab in 10 μL volume. The MF was dissolved in 95% ethanol and diluted with crab ringer so that the final concentration of ethanol was 10%. Reddy (1980) found that the hemolymph volume (mL) of *Oziotelphusa* is 27% of body mass. Assuming that the crab used in these experiments had 7.2–7.5 mL of hemolymph, then the final concentration of methyl farnesoate injected will be approximately 2 ng/mL hemolymph, which is the physiological concentration (Tobe *et al.*, 1989). Injections were given on the first, seventh, fourteenth, and twenty-first day, and the crabs were sacrificed on day 28. No deaths occurred in the control or in the experimental groups.

The crabs were immobilized by chilling on ice for 10 min. The body weight of the crabs was determined. The reproductive organs were isolated, immediately placed in ice-cold crustacean physiological saline (Van Harreveld, 1936) to scrape off adhesive tissue. The organs were removed from the saline and lightly blotted with paper towels, and weighed wet on an electronic balance. The reproductive stages in the crabs were identified according to Reddy and Ramamurthi (1998). In brief, oögonial proliferation and ovarian differentiation takes place in *O. senex senex* when the ovary is translucent to opaque white

(previtellogenic stage). During vitellogenesis, the colour of the ovary changes from pale yellow (vitellogenic stage I) to orange (vitellogenic stage II), and then it becomes brown (vitellogenic stage III) to almost dark brown prior to spawning. Maturation of the ovary also includes an increase in size of ovary as the oocytes proliferate and increase in oocyte diameter because of yolk deposition. The premolt stages were determined using changes in setal development in the mastigobranch of the third maxilliped as described for this crab (Reddy, 1991; Kishori and Reddy, 2004).

The gonad indices were determined using the standard formula:

$$\text{Gonad index} = \frac{\text{Wet weight of the gonad}}{\text{Wet weight of the crab}} \times 100.$$

After 24 h of fixation in aqueous Bouin's solution, the reproductive tissues were dehydrated through an alcohol series. The tissues were cleared in xylene and then embedded in paraffin wax (m.p. $56^\circ\text{--}58^\circ\text{C}$). Sections 7- μm thick were cut and stained with hematoxylin followed by eosin. The diameters of 25 randomly chosen oocytes/testicular follicles were measured using an ocular micrometer on a microscope.

The significance of differences between the means was determined by using one-way ANOVA followed by Student-Newman-Keuls test.

RESULTS

Effect of MF on Molting

Injection of MF results in recruitment of greater number of crabs into premolt stage (males 100%; females 80%). Among them, six (24%) males and four (16%) females had molted after the fourth dose (Tables 1, 2). The crabs that served as concurrent controls were in the intermolt stage (Tables 1, 2).

Effect of MF on Reproduction

The oocytes of the initial control crabs were in the previtellogenic stage, whereas by the end of the experiment (28 days) the oocytes of the concurrent control crabs had enlarged slightly (5.18%) but not significantly from control oocytes showing that only a small amount of ovarian growth had occurred during the experimental period in the crab. Injection of MF significantly ($P < 0.0001$) increased the ovarian index (1304.35%) and mean oocyte diameter (236.37%) when compared with the corresponding values for the control (Table 1). Dark brown-coloured ovaries were observed in 56% of crabs, whereas in the remaining crabs, the ovaries were in different stages of development (pale yellow, 32%; orange, 12%) after methyl farnesoate injection.

In male crabs, administration of MF resulted in a significant ($P < 0.0001$) increase in testicular index (38.00%) and mean testicular follicle diameter (25.72%) when compared with the initial control crabs. No significant change was observed in the concurrent control crabs when compared with the initial control crabs (Table 2).

DISCUSSION

Even though there is a large body of information available for the presence of methyl farnesoate in crustaceans, relatively few data have been provided concerning the physiological roles of methyl farnesoate. The possible roles of methyl farnesoate in regulating crustacean reproduction appear to have received more attention than any other physiological processes (Hinsch, 1980; Brody and Chang, 1989; Laufer, 1992; Laufer *et al.*, 1986, 1987b, 1997, 1998; Sagi *et al.*, 1993; Wilder *et al.*, 1995; Reddy and Ramamurthi, 1998; Kalavathy *et al.*, 1999). The evidence for the role of methyl farnesoate in the regulation of crustacean molting is also increasing (Yudin *et al.*, 1980; Chang, 1993; Tamone and Chang, 1993; Abdu *et al.*, 2001; Nagaraju *et al.*, 2004).

In the freshwater crab, *Oziotelphusa senex senex*, molting and reproduction are not sequential. The crab completes the entire reproductive process in the intermolt stage of the molt cycle (Reddy *et al.*, 1985). Because these processes are highly energy demanding, the animal may either molt or reproduce depending on complex external and internal factors. In nature, gonad growth was not observed in premolt crabs. In the present study, some of the crabs (females

Table 2. Effect of injection of methyl farnesoate on molting and reproduction in male crab, *Oziotelphusa senex senex*.

	Testicular index	Testicular follicle diameter (μm)	Molt stage
Control ($n = 25$)	0.7 ± 0.2	37.84 ± 3.65	Intermolt
Concurrent control ($n = 25$)	0.73 ± 0.34 (4.29)	38.24 ± 2.64 (1.057)	Intermolt (100)
MF injected ($n = 25$)	0.966 ± 0.07 (38.00)	47.573 ± 3.75 (25.72)	Premolt (76) Molted (24)
F ratio	6.767	48.015	
P value	0.0024	< 0.0001	

Values are mean $\pm$ S.D.

Values in parentheses are percentage change from control crabs.

64%; males 76%) not only entered into the premolt stage but also exhibited accelerated gonad growth after MF injection. The MF injected crabs, which did not enter into the premolt stage (20%), exhibited not only an increase in ovary size but also changes in colour (dark brown, 16%; orange, 4%). Comparison of data for the initial control crabs and the concurrent control crabs shows that the gonads were slowly developing during the course of these experiments. The crabs injected with methyl farnesoate showed greater gonad growth than did the concurrent control crabs, as is evident from the greater gonad indices, oocyte diameters, and testicular diameters. Administration of methyl farnesoate resulted in recruitment of a greater number of crabs into the premolt stage. Whether MF is directly involved in the regulation of molting or acts by stimulating the ecdysteroid production by the Y-organs in the crab *O. senex senex* remains to be studied.

In crustaceans, reproduction is regulated by two neuropeptide hormones—gonad inhibiting hormone synthesized and secreted from X-organ–sinus gland complex in the eyestalks and gonad stimulating hormone from brain and thoracic ganglia. In the present study, the stimulatory action of MF on gonad development could have been due to stimulation of gonad stimulating hormone synthesis and release or inhibition of release of gonad inhibiting hormone or direct action of MF on gonads. It could be that all the three hormones (viz., GSH, GIH, and MF) play a very important role in regulating reproduction in crustaceans. Because MF stimulated gonad development in the crab, the possibility of using MF to stimulate reproduction of other aquaculturally important crustaceans is

worth investing and is in progress in our laboratory.

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